1)

$$\left(\mathbf{r}-\begin{pmatrix}-1\\8\\2\end{pmatrix}\right)\cdot\begin{pmatrix}-2\\2\\1\end{pmatrix}=0$$

2)

$$\cos\theta=\frac{\begin{pmatrix}2\\0\\1\end{pmatrix}\cdot\begin{pmatrix}6\\-1\\2\end{pmatrix}}{\sqrt{2^2+0^2+1^2}\times\sqrt{6^2+(-1)^2+2^2}}$$

$$\Rightarrow\cos\theta=\frac{2\times6+0\times(-1)+1\times2}{\sqrt{5}\times\sqrt{41}}$$

$$\Rightarrow\cos\theta=\frac{14}{\sqrt{205}}\Rightarrow\theta=12.094\ldots°$$

Acute angle is 12.1° (1 DP)

3)

$$\left(\begin{pmatrix}3\\\frac{8}{5}\\7\end{pmatrix}-\begin{pmatrix}5\\2\\-3\end{pmatrix}\right)\cdot\begin{pmatrix}-6\\5\\-1\end{pmatrix}=\begin{pmatrix}-2\\-\frac{2}{5}\\10\end{pmatrix}\cdot\begin{pmatrix}-6\\5\\-1\end{pmatrix}$$

$$=(-2)\times(-6)+\left(-\frac{2}{5}\right)\times5+10\times(-1)$$

$$=12-2-10=0$$

So $\left(3,\frac{8}{5},7\right)$ lies on the plane Π

4)

$$\begin{pmatrix}2\\-5\\-4\end{pmatrix}\cdot\begin{pmatrix}-9\\8\\-16\end{pmatrix}=2\times(-9)+(-5)\times8+(-4)\times(-16)=-18-40+64=6$$

So $(2,-5,-4)$ lies on the plane Π

5) a)

$$\mathbf{r}\cdot\mathbf{i}=2\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}=2\Rightarrow x=2$$

5) b)

$$\mathbf{r}\cdot\mathbf{j}=0\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix}=0\Rightarrow y=0$$

5) c)

$$\mathbf{r}\cdot\mathbf{k}=-3\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}=-3\Rightarrow z=-3$$

5) d)

$$\mathbf{r}\cdot(3\mathbf{i}+2\mathbf{j}-3\mathbf{k})=9\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}3\\2\\-3\end{pmatrix}=9\Rightarrow 3x+2y-3z=9$$

5) e)

$$\mathbf{r}\cdot(10\mathbf{i}-50\mathbf{j}+25\mathbf{k})=30\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}10\\-50\\25\end{pmatrix}=30\Rightarrow 10x-50y+25z=30$$

$$\Rightarrow 2x-10y+5z=6$$

6)

The plane has equation $2x-3y-10z=d$

$d=2(3)-3(-6)-10(8)=-56$

$\therefore 2x-3y-10z=-56$

$$\Rightarrow\mathbf{r}\cdot\begin{pmatrix}2\\-3\\-10\end{pmatrix}=-56$$

7) a)

$$\cos\theta=\frac{\begin{pmatrix}-1\\-2\\4\end{pmatrix}\cdot\begin{pmatrix}3\\1\\10\end{pmatrix}}{\sqrt{(-1)^2+(-2)^2+4^2}\times\sqrt{3^2+1^2+10^2}}$$

$$\Rightarrow\cos\theta=\frac{(-1)\times3+(-2)\times1+4\times10}{\sqrt{21}\times\sqrt{110}}$$

$$\Rightarrow\cos\theta=\frac{35}{\sqrt{2310}}\Rightarrow\theta=43.262\ldots°$$

Acute angle is 43.3° (1 DP)

7) b)

$$\cos\theta = \frac{\begin{pmatrix}-5\\-1\\-9\end{pmatrix}\cdot\begin{pmatrix}16\\2\\-5\end{pmatrix}}{\sqrt{(-5)^2+(-1)^2+(-9)^2}\times\sqrt{16^2+2^2+(-5)^2}}$$

$$\Rightarrow \cos\theta = \frac{(-5)\times16+(-1)\times2+(-9)\times(-5)}{\sqrt{107}\times\sqrt{285}}$$

$$\Rightarrow \cos\theta = -\frac{37}{\sqrt{30495}} \Rightarrow \theta = 102.232\ldots°$$

Acute angle is $180° - 102.232\ldots° = 77.8°$ (1 DP)

8)

$$\cos\theta = \frac{\begin{pmatrix}3\\-2\\-10\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{3^2+(-2)^2+(-10)^2}\times\sqrt{0^2+0^2+1^2}}$$

$$\Rightarrow \cos\theta = \frac{3\times0+(-2)\times0+(-10)\times1}{\sqrt{113}\times\sqrt{1}}$$

$$\Rightarrow \cos\theta = -\frac{10}{\sqrt{113}} \Rightarrow \theta = 160.172\ldots°$$

Acute angle is $180° - 160.172\ldots° = 19.8°$ (1 DP)

9) a)

$$\begin{pmatrix}2\\3\\1\end{pmatrix}\cdot\begin{pmatrix}8\\-1\\1\end{pmatrix} = 2\times8+3\times(-1)+1\times1 = 14$$

$$\Pi_1: \mathbf{r}\cdot\begin{pmatrix}8\\-1\\1\end{pmatrix} = 14$$

9) b)

$$\begin{pmatrix}3\\9\\2\end{pmatrix}\cdot\begin{pmatrix}1\\-3\\-2\end{pmatrix} = 3\times1+9\times(-3)+2\times(-2) = -28$$

$$\Pi_2: \mathbf{r}\cdot\begin{pmatrix}1\\-3\\-2\end{pmatrix} = -28$$

9) c)

$$\cos\theta = \frac{\begin{pmatrix}8\\-1\\1\end{pmatrix}\cdot\begin{pmatrix}1\\-3\\-2\end{pmatrix}}{\sqrt{8^2+(-1)^2+1^2}\times\sqrt{1^2+(-3)^2+(-2)^2}}$$

$$= \frac{8\times1+(-1)\times(-3)+1\times(-2)}{\sqrt{66}\times\sqrt{14}}$$

$$= \frac{9}{\sqrt{924}} = \frac{3\sqrt{231}}{154}$$

10) a)

$$\begin{pmatrix}800t^2+540t-99\\300-20t\\800t^2+620t-133\end{pmatrix}\cdot\begin{pmatrix}1\\0\\-1\end{pmatrix}=0$$

$$\Rightarrow (800t^2+540t-99)\times1+(300-20t)\times0+(800t^2+620t-133)\times(-1)=0$$

$$\Rightarrow 800t^2+540t-99-800t^2-620t+133=0$$

$$\Rightarrow -80t+34=0 \Rightarrow t=\frac{17}{40}=0.425 \text{ seconds}$$

10) b)

$$\mathbf{v}|_{t=\frac{17}{40}} = \begin{pmatrix}1600\times\frac{17}{40}+540\\-20\times\frac{17}{40}\\1600\times\frac{17}{40}+620\end{pmatrix} = \begin{pmatrix}1220\\-\frac{17}{2}\\1300\end{pmatrix}$$

10) c)

$$\cos\theta = \frac{\begin{pmatrix}1220\\-\frac{17}{2}\\1300\end{pmatrix}\cdot\begin{pmatrix}1\\0\\-1\end{pmatrix}}{\sqrt{1220^2+\left(-\frac{17}{2}\right)^2+1300^2}\times\sqrt{1^2+0^2+(-1)^2}}$$

$$\Rightarrow \cos\theta = \frac{1220\times1+\left(-\frac{17}{2}\right)\times0+1300\times(-1)}{\sqrt{3178472.25}\times\sqrt{2}}$$

$$\Rightarrow \cos\theta = -\frac{80}{\sqrt{6356944.5}} \Rightarrow \theta = 91.818\ldots°$$

So the acute angle between the particle and the normal vector to the plane is $180° - 91.818\ldots° = 88.181\ldots°$

So the acute angle between the particle and the plane is $90° - 88.181\ldots° = 1.8°$ (1 DP)

11) a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 300 \\ 50 \\ -150 \end{pmatrix} - \begin{pmatrix} 150 \\ 0 \\ -200 \end{pmatrix} = \begin{pmatrix} 150 \\ 50 \\ 50 \end{pmatrix}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 400 \\ 150 \\ -300 \end{pmatrix} - \begin{pmatrix} 150 \\ 0 \\ -200 \end{pmatrix} = \begin{pmatrix} 250 \\ 150 \\ -100 \end{pmatrix}$

$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 150 \\ 50 \\ 50 \end{pmatrix} \times \begin{pmatrix} 250 \\ 150 \\ -100 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 150 & 50 & 50 \\ 250 & 150 & -100 \end{vmatrix}$

$= (50 \times (-100) - 50 \times 150)\mathbf{i} - (150 \times (-100) - 50 \times 250)\mathbf{j} + (150 \times 150 - 50 \times 250)\mathbf{k}$

$= -12500\mathbf{i} + 27500\mathbf{j} + 10000\mathbf{k}$

The plane has equation $-12500x + 27500y + 10000z = d$

$d = -12500(150) + 27500(0) + 10000(-200) = -3875000$

$\therefore -12500x + 27500y + 10000z = -3875000$

$\Rightarrow -5x + 11y + 4z = -1550$

So Π: $\mathbf{r} \cdot \begin{pmatrix} -5 \\ 11 \\ 4 \end{pmatrix} = -1550$

11) b)

$\cos\theta = \frac{\begin{pmatrix} -5 \\ 11 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}{\sqrt{(-5)^2+11^2+4^2} \times \sqrt{0^2+0^2+1^2}}$

$\Rightarrow \cos\theta = \frac{(-5)\times 0 + 11 \times 0 + 4 \times 1}{\sqrt{162} \times \sqrt{1}}$

$\Rightarrow \cos\theta = \frac{4}{\sqrt{162}} = \frac{2\sqrt{2}}{9} \Rightarrow \theta = 71.683\ldots°$

Acute angle is 71.7° (1 DP)